

NOTE TAKING APP

A Full-Stack CRUD Web Application

Project Report

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1. Introduction

The Note Taking App is a full-stack web application that allows registered users to create, view, edit, organize, search, pin, and delete personal notes. It provides each user with a private, authenticated workspace where notes are securely tied to their own account and inaccessible to anyone else. The application was developed as an academic CRUD (Create, Read, Update, Delete) project to demonstrate practical skills in full-stack web development, including session-based authentication, RESTful API design, and database modeling.

The application was built to closely mirror the functionality of popular note-taking tools such as Google Keep and Apple Notes, while remaining scoped appropriately for a beginner-to-intermediate academic project. It was developed by implementing two mandatory core concepts specified for this project: user authentication and full CRUD functionality for notes.

1.1 Background

In everyday academic and professional life, people frequently need a fast and reliable way to capture short pieces of information — reminders, ideas, to-do items, or reference material — without the overhead of a full document editor. While many note-taking tools exist commercially, building one from scratch offers deep insight into how modern web applications manage user accounts, protect private data, and perform structured data operations against a database. This project addresses that need by implementing a lightweight, secure, and easy-to-use digital note manager.

1.2 Problem Statement

Users need a simple, secure, and accessible way to create, organize, retrieve, edit, and remove personal notes without worrying about their data being visible to or editable by anyone else. The system must ensure that only the authenticated owner of a note can view or modify it, and that user credentials such as passwords are never stored or exposed in plain text.

1.3 Objectives

- To develop a secure user authentication system using session-based login with hashed password storage.
- To allow users to create, view, edit, and delete personal notes through a RESTful API.
- To implement strict data ownership so that a user can only access their own notes.
- To provide note organization features such as search and pinning for improved usability.
- To design a clean, minimal, and professional user interface using React and Tailwind CSS.

1.4 Scope and Limitations

Scope:

- User registration, login, logout, and session-based authentication.
- Full CRUD functionality for personal notes (create, read, update, delete).
- Note ownership enforcement — users cannot access notes belonging to other accounts.
- Search functionality across note titles and content.
- Pin/unpin functionality to highlight important notes.
- Responsive, styled user interface built with React, Vite, and Tailwind CSS.

Limitations:

- The application does not currently support note sharing or collaboration between users.
- Tag-based categorization exists in the underlying data model but is not yet exposed in the user interface.
- The application does not include file or image attachments within notes.
- Password reset via email is not implemented, as email delivery was outside the project's scope.

2. Tech Stack

The following technologies were used to build the Note Taking App, as specified by the project requirements:

Layer	Technology	Reason
Frontend	React + Vite	Component-based frontend with fast development builds
Styling	Tailwind CSS	Utility-based styling for a clean, consistent UI
Icons	Lucide-react	Consistent, lightweight UI icons
Backend	Node.js + Express	REST API development
Database	MongoDB	Document-based storage suited to flexible note data
ODM	Mongoose	MongoDB schema and model management
Authentication	express-session	Session-based authentication (per project requirement)
Session Store	connect-mongo	Persistent session storage in MongoDB

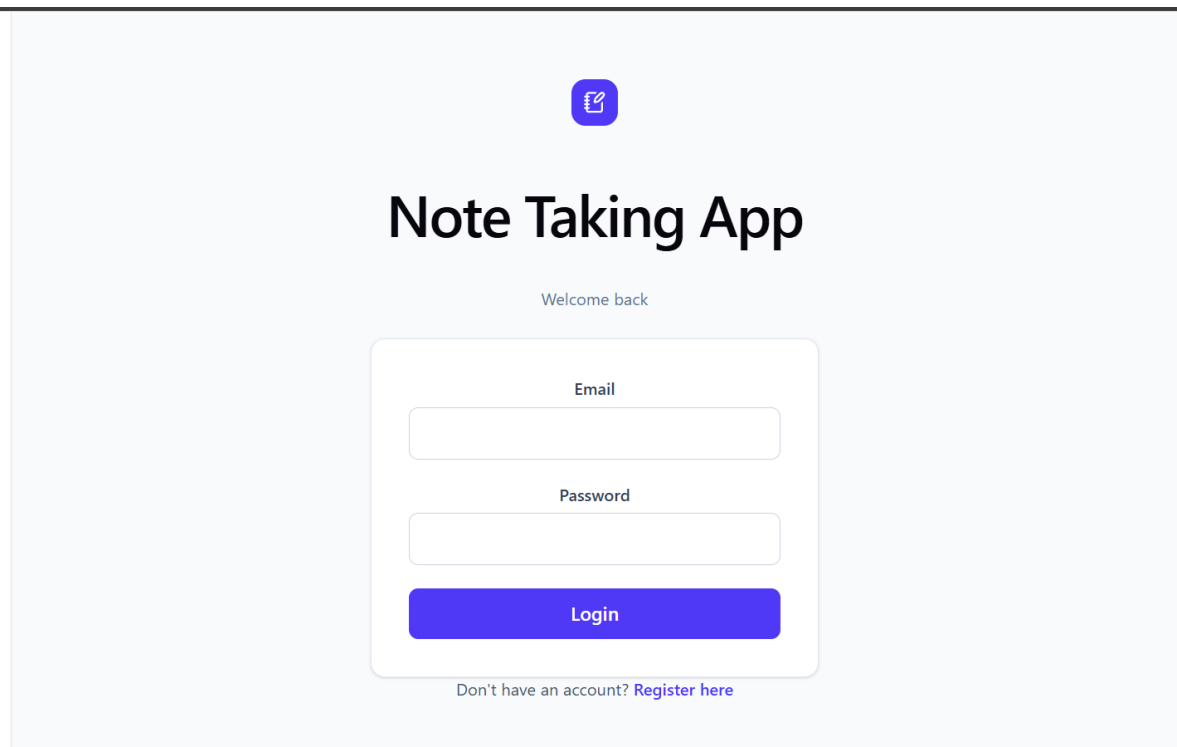
Layer	Technology	Reason
Security	bcryptjs	Password hashing (pure-JavaScript bcrypt implementation)
Version Control	Git + GitHub	Source-code management
Testing	Postman	API endpoint testing

Note: bcryptjs was used in place of the native bcrypt package. Both implement the same bcrypt hashing algorithm and expose an identical API; bcryptjs is a pure-JavaScript implementation, which avoids the native compilation step required by bcrypt and simplifies installation across different development environments.

3. Core Modules

3.1 Main UI Modules

Login Page



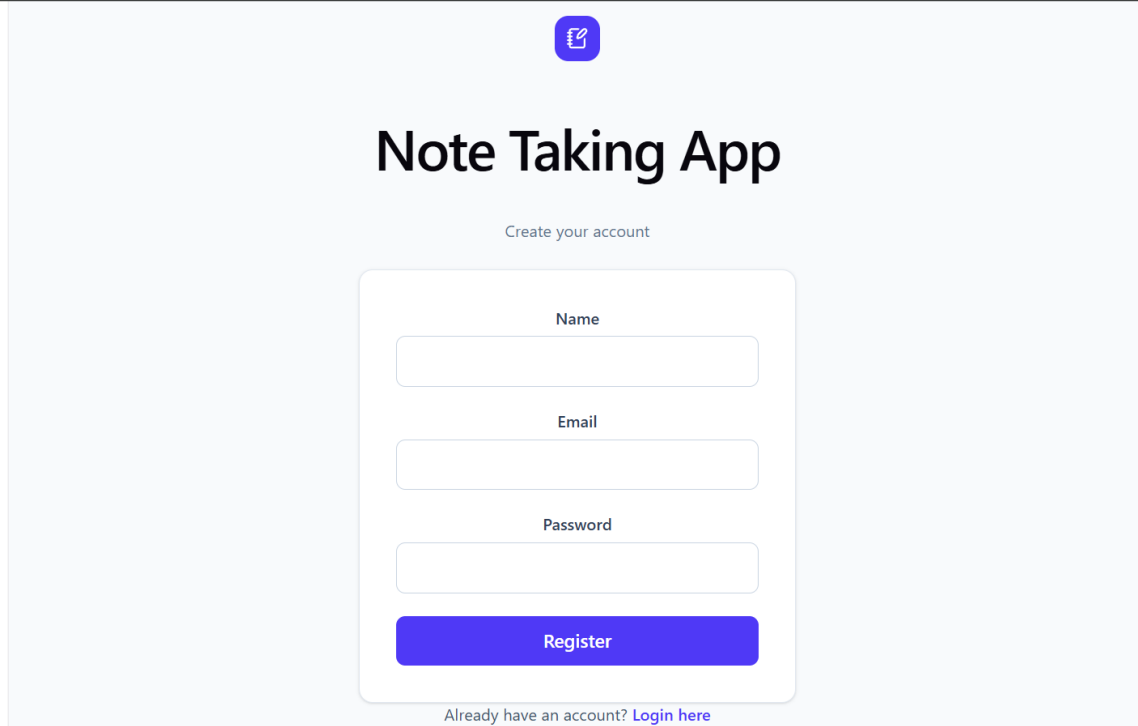
The screenshot shows the login page of the 'Note Taking App'. At the top center is a purple square icon with a white notepad and pencil. Below it, the title 'Note Taking App' is displayed in a large, bold, black font. Under the title, the text 'Welcome back' appears in a smaller, grey font. The main content is a white rounded rectangle containing three input fields: 'Email' and 'Password' (both with placeholder text and light blue borders), and a solid purple 'Login' button. At the bottom of this white box, there is a link that says 'Don't have an account? Register here' in a small, grey font, with 'Register here' being a blue hyperlink.

Purpose: Allows an existing user to authenticate using their email and password.

Main elements: Email input, password input, login button, link to the registration page.

User actions: Enter credentials and log in; on success, the user is redirected to the Dashboard and a session cookie is issued.

Register Page



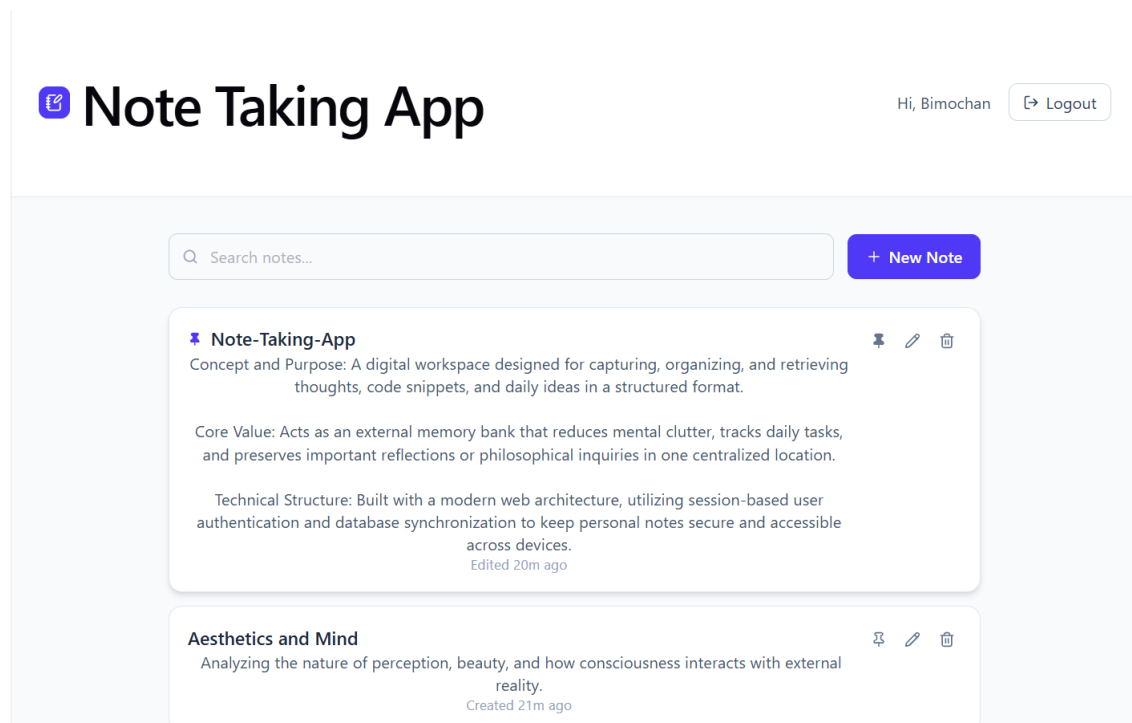
The register page features a light blue background. At the top center is a purple icon of a notepad with a pencil. Below it is the title "Note Taking App" in a large, bold, black font. Under the title is the text "Create your account" in a smaller, gray font. In the center is a white rounded rectangle containing three input fields labeled "Name", "Email", and "Password". Below these fields is a purple "Register" button. At the bottom of the white box is the text "Already have an account? [Login here](#)".

Purpose: Allows a new user to create an account.

Main elements: Name, email, and password inputs; register button; link to the login page.

User actions: Submit registration details to create an account; the password is hashed before storage and the user is automatically logged in.

Dashboard / Main Notes Page



The dashboard page has a light blue background. At the top left is a purple notepad icon followed by the title "Note Taking App". At the top right, it says "Hi, Bimochan" and a "Logout" button. Below the title is a search bar with the placeholder "Search notes..." and a purple "+ New Note" button. The main content area contains two note cards. The first card is titled "Note-Taking-App" and contains text about its concept, purpose, core value, and technical structure. The second card is titled "Aesthetics and Mind" and contains text about analyzing perception and beauty. Both cards have icons for pinning, editing, and deleting.

Purpose: The main authenticated workspace where users manage their notes.

Main elements: Header with user's name and logout button, search bar, "New Note" button, list of note cards showing title, content preview, pin status, and last-updated time.

User actions: Create, search, pin/unpin, edit, and delete notes.

Create / Edit Note Form

The screenshot displays the 'Note Taking App' interface. At the top, there's a header with a purple icon, the title 'Note Taking App', the user name 'Hi, Bimochan', and a 'Logout' button. Below the header is a search bar with the placeholder 'Search notes...' and a 'Cancel' button. The main form area contains two input fields: 'Title' and 'Content'. At the bottom of the form are 'Cancel' and 'Add Note' buttons. Below the form, a list of notes is shown. The first note is titled 'Note-Taking-App' and has a description: 'Concept and Purpose: A digital workspace designed for capturing, organizing, and retrieving thoughts, code snippets, and daily ideas in a structured format.' It also has icons for pinning, editing, and deleting.

Purpose: A form used both to create a new note and to edit an existing one.

Main elements: Title input, content textarea, save/cancel buttons.

User actions: Enter or update note details and submit to save changes.

Delete Confirmation

The screenshot shows a note card titled 'Note-Taking-App' with the same description as in the previous image. The note card has icons for pinning, editing, and deleting. A 'Delete' button is visible in the top right corner of the note card. The note card also includes a 'Core Value' section: 'Acts as an external memory bank that reduces mental clutter, tracks daily tasks, and preserves important reflections or philosophical inquiries in one centralized location.' and a 'Technical Structure' section: 'Built with a modern web architecture, utilizing session-based user authentication and database synchronization to keep personal notes secure and accessible across devices.' The note was 'Edited 22m ago'.

Purpose: Prevents accidental note deletion.

Main elements: A confirmation prompt triggered before a delete request is sent.

User actions: Confirm or cancel the deletion of a note.

3.2 Main API Endpoints

Method	Endpoint	Description	Example Use
POST	/api/auth/signup	Creates a new user account	Registering a new user
POST	/api/auth/login	Authenticates a user and starts a session	Logging in
POST	/api/auth/logout	Destroys the current session	Logging out
GET	/api/auth/me	Returns the currently authenticated user	Checking login state on app load
POST	/api/notes	Creates a new note for the logged-in user	Adding a note
GET	/api/notes	Retrieves all notes belonging to the user	Loading the notes list
GET	/api/notes/:id	Retrieves a single note by ID	Opening a note
PUT	/api/notes/:id	Updates an existing note	Editing a note or toggling pin status
DELETE	/api/notes/:id	Deletes a note	Removing a note

All note-related endpoints are protected by an authentication middleware that checks for a valid session before allowing the request to proceed, and every note operation additionally verifies that the requested note belongs to the currently logged-in user before returning or modifying it.

4. Conclusion

The Note Taking App successfully demonstrates the core principles of full-stack web development: secure user authentication, RESTful CRUD operations, and strict data ownership enforcement. Through this project, practical experience was gained in building a session-based authentication system with hashed password storage, designing a REST API

with Express and MongoDB, and building a responsive, professional-looking user interface with React and Tailwind CSS.

The project met its core objectives: users can securely register and log in, manage their own private notes through full CRUD functionality, search and organize notes via pinning, and are protected from accessing or modifying notes that do not belong to them. While some optional features such as tagging and note sharing remain unimplemented, the application fulfills all mandatory requirements specified for the project and provides a solid foundation for future enhancement.

5. References

- Express.js Documentation — <https://expressjs.com/>
- Mongoose Documentation — <https://mongoosejs.com/docs/>
- MongoDB Documentation — <https://www.mongodb.com/docs/>
- React Documentation — <https://react.dev/>
- Vite Documentation — <https://vitejs.dev/>
- Tailwind CSS Documentation — <https://tailwindcss.com/docs>
- express-session (npm) — <https://www.npmjs.com/package/express-session>
- connect-mongo (npm) — <https://www.npmjs.com/package/connect-mongo>
- bcryptjs (npm) — <https://www.npmjs.com/package/bcryptjs>